On some quantitative aspects of categorical Lagrangian topology

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DISS. ETH NO. 31947

ON SOME QUANTITATIVE ASPECTS OF CATEGORICAL LAGRANGIAN TOPOLOGY

A thesis submitted to attain the degree of

DOCTOR OF SCIENCES
(Dr. sc. ETH Zürich)

presented by

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born on 30.11.1997

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2026

Sommario

Questa tesi di dottorato consiste principalmente in una presentazione organizzata e coerente di due articoli di ricerca [**Amb25**] e [**ABC26**] co-firmati dall'autore. Il lavoro è dedicato alla definizione della categoria di Fukaya derivata persistente, un raffinamento della nozione di categoria di Fukaya derivata —ormai classica della topologia simplettica — e allo studio di alcune delle sue proprietà.

La tesi è suddivisa in tre parti. Nella prima parte vengono introdotti gli strumenti algebrici necessari per la definitione e lo studio delle categorie di Fukaya derivate persistenti, in particolare le categorie A_∞ filtrate, le categorie triangolate persistenti (TPCs) e la nozione astratta di approssimabilità metrica categorica, interpretabile sia come una generalizzazione della nozione metrica di spazio totalmente limitato, sia come una versione persistente del concetto di generazione triangolare nelle categorie triangolate.
Nella seconda parte viene sviluppato un modello di categoria di Fukaya filtrata per alcune classi di varietà simplettiche. Questo passaggio preliminare è necessario per la definizione di categorie di Fukaya derivate persistenti. Gli strumenti principali sono: (i) un modello "cluster" per la categoria di Fukaya e (ii) la costruzione di una classe adeguata di perturbazioni hamiltoniane.
Nella terza parte si dimostra che alcune classi di sottovarietà lagrangiane, dotate della metrica spettrale lagrangiana, sono categoricamente approssimabili in opportune categorie di Fukaya derivate persistenti. Si mostra inoltre che, in alcuni casi, tali classi sono approssimabili ma non totalmente limitate.

Abstract

This doctoral thesis consists in an organised and coherent presentation of two articles [Amb25] and [ABC26] co-authored by the present author. The thesis is devoted to the definition of the persistent derived Fukaya category, which refines the classical notion of the derived Fukaya category in symplectic topology, and to the study of some of its main features.

The thesis consists of three parts. In the first part we introduce all the necessary algebraic material needed to discuss persistence derived Fukaya categories and categorical metric approximability, that is, filtered A_∞-categories, TPCs and the abstract notion of categorical metric approximability, which can be seen as a weaker version of total boundedness as well as a persistence version of triangular generation in triangulated categories.
In the second part we develop a model for a filtered Fukaya category of certain symplectic manifolds. This is a necessary step in order to define TPC refinements of derived Fukaya categories. The main tools are: (i) a cluster model for the Fukaya category and (ii) the construction of an adequate class of Hamiltonian perturbation data.
In the third part we show that certain classes of Lagrangian submanifolds endowed with the Lagrangian spectral metric are categorically metric approximable in appropriate persistence derived Fukaya categories. We show that, in some cases, these classes are approximable but not totally bounded.

che nerd mado

Giulio Bozzini, March 2026

Acknowledgments

This thesis marks the end of a ten-year period of my life. Its completion coincides with the end of my stay in Zürich and represents a clear break with my previous life. While it is the result of my academic journey at ETH Zürich, I see it more broadly—and perhaps more romantically—as the outcome of years of experiences, struggles, journeys, questions, successes and failures, beginnings and ends. In these pages, study, time, relationships and obessessions come together, both in serie and in parallel, and their interplay has given shape what's contained in this thesis.

I would like to start by thanking my advisor Paul Biran. I learned a lot of mathematics from him, and I am very grateful for the advices I received and for the support he showed me during difficult times in my doctoral path. I am quite convinced that other than a great mathematician, I was given the opportunity of working with a very nice human being. Next, I want to deeply thank Octav Cornea for believing in me, appreciating my work and for being for me a good source of motivation. I learned a lot by collaborating with them.

Contents

CHAPTER 1

Introduction

This thesis aims to lay the foundations for the study of persistence structures in symplectic topology from a categorical perspective. It is primarily based on material developed in [**Amb25**] and [**ABC26**].

The use of persistent homology methods in symplectic topology dates back to [**PS16**] and has since become a well-established direction of research. The key observation is that Floer complexes often admit a natural filtration induced by the action functional. Independently, categorical structures in symplectic topology have been studied at least since Fukaya's pioneering work [**Fuk97**] and play a central role in Kontsevich's homological mirror symmetry conjecture [**Kon95**].

In this thesis we try to bridge the gap between these two directions of study. To this aim, work on both algebraic and geometric foundations of such a theory needs to be undertaken.

On the algebraic side, we build on the work of Biran–Cornea–Zhang [**BCZ24**] on triangulated persistence categories (TPCs) and introduce the notion of *categorical metric approximability* in Chapter **??**. This concept is inspired by Turing's notion of approximability for metric groups [**Tur38**]. Categorical metric approximability for Fukaya categories constitutes the main focus of the joint work [**ABC26**] with Biran and Cornea.

On the geometric side, we develop a model for the Fukaya category of (certain) symplectic manifolds that is compatible in a precise sense with the filtration coming from the action functional. This is the content of Chapter **??** and is based on the publication [**Amb25**].

Summarizing:

- Chapter 2 contains material on filtered *dg* and A_∞-categories, persistence categories, TPCs and (categorical metric) approximability, and is an extension of material contained in [**Amb25**] and [**ABC26**].
- Chapter **??** contains an exposition of the construction of the filtered Fukaya category from [**Amb25**], with some extensions.
- Chapter **??** contains the proof of the approximability results for spaces of Lagrangian submanifolds from [**ABC26**].

In the remainder of this chapter, we describe in greater detail the material covered in Chapters 2, **??**, and **??**, and explain the overall organization of the thesis.

1.1. Chapter 2: Algebraic preliminaries

We briefly summarize the content of Chapter 2.

In this chapter we first introduce filtered *dg*-categories as enriched categories over filtered chain complexes. We then introduce filtered A_∞-categories, and some related machinery, such as filtered A_∞-functors with linear deviation and persistence Hochschild homology. Given a filtered A_∞-category $\mathcal{A}$, we explain how to obtain a pre-triangulated filtered A_∞-category $\mathcal{A}^\triangle$ via filtered A_∞-modules and filtered twisted complexes.

We discuss the basic properties of persistence categories, originally introduced in [**BCZ24**]. In this thesis, we adopt a slighlty different viewpoint and present persistence categories as categories enriched over the symmetric monoidal category of persistence modules. We hope that, in future work, this viewpoint might be helpful to shed light on metric properties of triangulated categories of interest in symplectic topology. In particular, the morphism spaces of a persistence category $\mathcal{C}$ is a persistence module $\hom_{\mathcal{C}}(L_0, L_1)$ for any two objects L_0 and L_1 of $\mathcal{C}$. It follows that a persistence category $\mathcal{C}$ naturally defines two other categories $\mathcal{C}^0$ and $\mathcal{C}^\infty$. Both $\mathcal{C}^0$ and $\mathcal{C}^\infty$ have the same objects as $\mathcal{C}$, but morphism spaces defined by

$$\hom_{\mathcal{C}^0}(L_0, L_1) := \hom_{\mathcal{C}}(L_0, L_1)^0 \text{ and } \hom_{\mathcal{C}^\infty}(L_0, L_1) := \hom_{\mathcal{C}}(L_0, L_1)^\infty$$

where the superscript 0 indicates the module lying at persistence level 0 in a persistence module, while the superscript ∞ indicates the terminal module of a persistence module.

A persistence category $\mathcal{C}$ naturally admits a pseudo-metric d_{int} on its objects. This metric is induces from the classical interleaving distance on persistence modules, which, we recall, are the basis of the enrichment of a persistence category. We will hence call d_{int} the "interleving distance of $\mathcal{C}$". Note that, since d_{int} comes from the interleaving distance, two objects L_0 and L_1 of $\mathcal{C}$ satisfy $d_{\text{int}}(L_0, L_1) < \infty$ if and only if they become isomorphic objects in $\mathcal{C}^\infty$.

One should think of persistence categories as quantitative refinements of (preadditive) categories. Given a category $\mathcal{D}$ it is sometimes of interest to find a persistence category $\mathcal{C}$ such that $\mathcal{D}$ is equivalent to $\mathcal{C}^\infty$. If this is the case, then one gets an interleaving distance on the objects of $\mathcal{D}$.

Note that persistence categories naturally appear as the homotopy category of filtered A_∞-categories, similarly to how persistence modules appear as homologies of filtered chain complexes.

The central categorical object of this chapter are triangulated persistence categories, a refinement of the classical notion of triangulated categories in the persistence setting, also introduced in [**BCZ24**]. A triangulated persistence category is, roughly said, a persistence category $\mathcal{C}$ such that the 0-level category $\mathcal{C}^0$ is a triangulated category in the classical sense, i.e. it comes with a translation endofunctor and a distinguished class of couple of morphisms, usually called "exact triangles". It easily follows by the properties of a TPC, that the terminal category $\mathcal{C}^\infty$ naturally admits the structure of triangulated category.

Similarly as in the case of persistence categories above, one should think of TPCs as enhancements of triangulated categories: let $\mathcal{T}$ be a triangulated category (usually of geometric origin), we say that $\mathcal{T}$ admits a TPC refinement if there is a TPC $\mathcal{C}$ such that $\mathcal{C}^\infty$ is triangulated equivalent to $\mathcal{T}$.

In TPCs, we can exploit the persistence structure to refine classical notions and invariants for triangulated category. Most importantly, the structure of a TPC allows for the definition of "categorical metric approximability" introduced in [**ABC26**]. We briefly sketch the definition here. First, we consider the intrinsic case, then we introduce the extrinsic definition.

Let $\mathcal{T}$ be a triangulated category and consider a triangular generator L of $\mathcal{T}$. Then, given any object L' of $\mathcal{T}$, there is an iterated cone C in $\mathcal{T}$, built from L, such that L' is isomorphic to $C_{L'}$ in $\mathcal{T}$. Assume that $\mathcal{T}$ admits a TPC refinement $\mathcal{C}$. Via algebraic manipulation, one can transform $C_{L'}$ into an iterated cone $\mathcal{C}^0_{L'}$ in $\mathcal{C}^0$, and we say that L ε-approximates L' if

$$d_{\mathrm{int}}(L', C^0_{L'}) \leq \varepsilon$$

for some $\varepsilon \geq 0$, where d_{int} is the interleaving distance on $\mathcal{C}$. We can ask ourselves the following question: do we get a better approximation, if instead of a single generator L we consider a family of generators $\mathcal{F}$ and iterated cones built from objects of $\mathcal{F}$? It turns out that this question is of interest in the case of Fukaya categories (see Chapter **??**).

This motivates the following two definitions, which are a rough version of the definitions appearing in §**??**, which are categorified versions of a notion of approximability for metric groups due to Turing [**Tur38**] (see the discussion in §**??**).

Definition 1.1.0.1. Let $\mathcal{T}$ be a triangulated category and $\mathcal{F}$ and $\mathcal{S}$ two families of objects in $\mathcal{T}$. We say that $\mathcal{F}$ approximates $\mathcal{S}$ if there is a TPC refinement $\mathcal{C}$ of $\mathcal{T}$ such that for any $\varepsilon > 0$ there is a "finite" subset $\mathcal{F}_\varepsilon \subset \mathcal{F}$ such that for any object L of $\mathcal{S}$ there is a finite iterated cone C_L in $\mathcal{C}^0$ built from objects of $\mathcal{F}_\varepsilon$ satisfying

$$d_{\mathrm{int}}(L, C_L) \leq \varepsilon.$$

We will often say that, in the situation above, the subset $\mathcal{S}$ of $\mathrm{Obj}(\mathcal{T})$ is approximable in the refinement $\mathcal{C}$ of $\mathcal{T}$. The extrinsic viewpoint is as follows.

Definition 1.1.0.2. Let $(\mathcal{L}, d)$ be a pseudo-metric space. We say that $(\mathcal{L}, d)$ is approximable if there is a triangulated category $\mathcal{T}$ such that

1. $\mathcal{L}$ can be identified with a subset $\mathcal{S}_\mathcal{L}$ of the objects of $\mathcal{T}$,
2. There is a TPC refinement $\mathcal{C}$ of $\mathcal{T}$ such that the restriction of the interleaving distance d_{int} on $\mathcal{C}$ to the subset $\mathcal{S}_\mathcal{L}$ coincides with d,
3. $\mathcal{S}_\mathcal{L}$ is approximable in $\mathcal{C}$.

We define a weaker notion than approximability, which we call "retract" approximability, and can be thought of as the counterpart of split-generation for triangulated categories in this quantitative context.

Remark 1.1.0.3. It is worth remarking at this point that there is another notion of approximability for triangulated categories, due to Neeman [**Nee22**].

Remark 1.1.0.4. a. We relate the concept of approximability in Definition 1.1.0.2 to the classical concept of total boundedness. See the introduction to [**ABC26**] for a more detailed exposition. We recall that a pseudo-metric subspace $(X, d_X) \subset (Y, d_Y)$ is totally bounded if for any $\varepsilon > 0$ there is a finite ε-net for X in Y, that is a finite set $X_\varepsilon \subset Y$ such that for any $x \in X$ there is $y \in X_\varepsilon$ such that $d_Y(x, y) < \varepsilon$. Approximability can be seen as a weak version of total boundedness in the sense that the ε-net $X_\varepsilon = \langle F_\varepsilon \rangle$ is only assumed to be a *finitely generated* (in a triangular sense) subcategory of a triangulated category.

b. Recall that a totally bounded pseudometric space is in particular bounded, and hence of finite diameter. The reverse implication does not hold. We are not aware neither of a relationship between TPC approximability and boundedness, neither of an example of pseudometrispace being approximable but not bounded.

Remark 1.1.0.5. The notion of approximability can be defined without appealing to the machinery of TPCs (see Definition **??**). However all the examples known to date make use of TPCs.

Along the way, in §**??**, we define the concept of system of A_∞-cateogories and TPCs with increasing accuracy, which is, at the moment, needed to deal with filtered Fukaya categories. Indeed, in our geometric situation in Chapters **??** and **??** one naturally obtains not a single filtered A_∞-category but a family of filtered A_∞-categories depending on auxiliary choices. This is formalized by a directed parameter space $(\mathcal{P}, \preceq, \nu)$, where $\nu(p) \geq 0$ measures the error and can be made arbitrarily small. A *system of filtered A_∞-categories with increasing accuracy* consists of categories $\{\mathcal{A}_p\}_{p \in \mathcal{P}}$ together with comparison A_∞-functors $\mathcal{H}_{p,q} \colon \mathcal{A}_p \to \mathcal{A}_q$ and $\mathcal{H}_{q,p}$ for all $p, q \in \mathcal{P}$, such that $\mathcal{H}_{p,q}$ preserves the filtration for $p \preceq q$. These functors are unique up to isomorphism in persistence homology, compatible with composition, and approximately inverse in the sense that the failure of $\mathcal{H}_{q,p} \circ \mathcal{H}_{p,q}$ to be the identity is controlled linearly by $|C\nu(p) - \nu(q)|$, $C \in [0, \infty)$. As $\nu(p) \to 0$ these comparisons become arbitrarily precise and all categories represent the same object up to homotopy. If, in addition, all comparison functors are unique up to 0-homotopy and their compositions are 0-homotopic to the direct comparisons, the system is called a *homotopy system*, which ensures that invariants such as persistence Hochschild homology are well defined at the level of the whole system. A closely analogous notion exists for TPC.

1.2. Chapter 3: Filtered Fukaya categories

We briefly summarize the content of Chapter **??**.

Let (X, ω) be a closed or convex at infinity symplectic manifold, and denote by $\mathcal{L}\mathrm{ag}^*$ a family of closed Lagrangian submanifolds of X belonging to some class $*$ (e.g. $* = m$ for monotone Lagrangians, $* = w\text{-}ex$ for weakly exact Lagrangians, $* = ex$ for exact Lagrangians). In this chapter of the thesis we will work with monotone Lagrangians, but our construction

works for weakly exact Lagrangians and exact Lagrangians in Lioville manifolds too.
We will refer to the model of the Fukaya category developed in Seidel's book [**Sei08**] as the *standard model for the Fukaya category of X* (see also e.g. [**BC13, BC14, BCS21**] for this model adapted to the monotone setting). In this standard model, morphism sets are Floer complexes $CF(L, L')$ where $L, L' \in \mathcal{L}\text{ag}^*$ and the A_∞-maps

$$\mu_d : CF(L_0, L_1) \otimes \cdots \otimes CF(L_{d-1}, L_d) \to CF(L_0, L_d)$$

are defined for any $d \geq 1$ and any tuple $(L_0, \ldots, L_d)$ of Lagrangians in $\mathcal{L}\text{ag}^*$ by counting Hamiltonian perturbed pseudoholomorphic polygons with Lagrangian boundary conditions, so-called Floer polygons. The representatives of homological units are constructed by counting Floer "1-gons". The action functional (defined in §**??**) induces an increasing real filtration on Floer complexes, that is a choice of subspaces $CF^{\leq \alpha}(L, L')$ for any $\alpha \in \mathbb{R}$ such that

$$CF^{\leq \alpha}(L, L') \subset CF^{\leq \beta}(L, L') \text{ for any } \alpha \leq \beta.$$

It is well-known that μ_1 preserves the above defined filtration, while in general higher order maps do not, and representatives of the units lie at positive filtration levels (see [**BCS21**]). Theorem **??** in this chapter, asserts that it is possible to construct a model for the Fukaya category admitting filtration-preserving A_∞-maps as well as units lying in filtration level ≤ 0. We show that there is a space $\mathcal{P}$ of perturbation data indexing a family of filtered Fukaya categories. Subsequently, we construct continuation A_∞-functors, an A_∞-extension of continuation chain maps, in §**??**, following [**Syl19**]. In Theorem **??**, we show that continuation functors are A_∞-functor with controlled linear deviation.

Using the algebraic notions developed in Chapter 2, the results contained in this chapter can be summarized as follows.

THEOREM 1.2.0.1. *There is a non-empty space $\mathcal{P}$ of perturbation data and, for any $p, q \in \mathcal{P}$, a non-empty space $\mathcal{J}_{p,q}$ of "continuation" data indexing a homotopy system of filtered and strictly unital A_∞-categories with increasing accuracy*

$$\widehat{\mathcal{F}uk}(\mathcal{L}ag^*) = \big(\mathcal{F}uk(\mathcal{L}ag^*; p)_{p \in \mathcal{P}}, (\mathcal{H}^h_{p,q} \colon \mathcal{F}uk(\mathcal{L}ag^*; p) \to \mathcal{F}uk(\mathcal{L}ag^*; q))_{p,q \in \mathcal{P}, h \in \mathcal{J}_{p,q}}\big)$$

where each A_∞-category $\mathcal{F}uk(\mathcal{L}ag^*; p)$, $p \in \mathcal{P}$, is quasi-equivalent to the standard model from [**Sei08**].

To prove Theorem 1.2.0.1 we have to tackle two problems: define A_∞-maps that do not shift filtration, and find represenatives of the unit lying at vanishing filtration levels. To solve the first problem we will inductively construct a class of perturbation data (à la [**Sei08**]) so that the associated Fukaya category is filtered. However, as we will argue in §**??**, it is not possible to find representatives of the units at zero filtration levels by using Seidel's standard model. We solve this problem by replacing self-Floer complexes with Morse complexes, whence the need for the so-called cluster complex.
The idea behind the construction of our perturbation data is roughly as follows: we pick "flat enough" Hamiltonian Floer data for any couple of Lagrangians in $\mathcal{L}\text{ag}^*$ and use coordinates

induced by strip-like ends on Floer polygons to interpolate between the Hamiltonian Floer datum and the zero function by a particular class of monotone homotopies. The precise construction is given in §**??**. In §**??** we argue that it is not possible to find representatives of the units at zero filtration levels by using the standard model for the Fukaya category. To cope with this fact, we use a Morse-bott, or "cluster", model for Fukaya categories, by defining a new A_∞-category whose self-morphisms spaces are pearl complexes, instead of Floer ones, and by considering "clusters" of pearly-Morse trees and Floer polygons. This model was first introduced in [**CL06**] and later outlined in [**BCZ24**] in the exact case; we work out more details of this model in §**??**.

Note that, as argued in [**BCZ24**], a filtered A_∞-category $\mathcal{A}$ induces a tringulated persistence category $PD(\mathcal{A})$ (see 2). Since, moreover, homotopy systems behave well when passing to the persistence derived level, Theorem 1.2.0.1 has the following corollary, which will be crucial for the results developed in Chapter **??**.

Corollary 1.2.0.2. *The homotopy system $\widehat{\mathcal{F}uk}(\mathcal{L}ag^*)$ induces a system of TPCs with increasing accuracy $PD(\widehat{\mathcal{F}uk}(\mathcal{L}ag^*))$.*

We remark that, by construction, the TPCs in the system $PD(\widehat{\mathcal{F}uk}(\mathcal{L}\mathrm{ag}^*))$ share a subset of objects, that is $\mathcal{L}\mathrm{ag}^*$. However, even though $\widehat{\mathcal{F}uk}(\mathcal{L}\mathrm{ag}^*)$ is a homotopy system, $PD(\widehat{\mathcal{F}uk}(\mathcal{L}\mathrm{ag}^*))$ does not have a coherent fixed base of objects (see Remark **??**).

1.3. Chapter 4: Approximability for Lagrangian submanifolds

We briefly summarize the content of Chapter **??**.

This chapter of the thesis contains the central three sections in [**ABC26**]. The main result is as follows.

Theorem 1.3.0.1. *The following three classes of Lagrangians $\mathcal{L}ag^*$ are approximable, as below.*

- *i. Closed, exact Lagrangians in unit cotangent disk bundles $M = D^*N$ for any closed manifold N (and any metric on N) are TPC approximable.*
- *ii. Equators on the 2-sphere, $M = S^2$, are TPC retract approximable.*
- *iii. Non-contractible Lagrangians on the 2-torus, $M = \mathbb{T}^2$, are TPC retract approximable.*

In all three cases TPC approximability takes place in appropriate TPC refinements of the derived Fukaya category and the relevant spaces of Lagrangian submanifolds $\mathcal{L}ag(M)$ are endowed with the metric structure given by the spectral metric d_γ in the cases i and ii and with a metric that restricts to the spectral metric on each Hamiltonian isotopy class at point iii.

The symplectic form on the cotangent disk bundles D^*N is the derivative of the Liouville form λ and the exactness of Lagrangians is understood with respect to λ. The ε-approximating families in the three cases consist of: fibers of the cotangent-disk bundle in the first case; great circles on the sphere connecting the north and south poles in the second; one latitude and a (finite) collection of longitudes on the torus in the third.

As explained in §1.1, approximability of a metric space can be viewed as a "triangular relaxation" of total boundedness. We prove that there are examples of spaces of Lagrangian endowed with d_γ that are approximable but not totally bounded.

COROLLARY 1.3.0.2. *The spaces $(\mathcal{L}ag(M), d_\gamma)$ are not totally bounded when $M = D^*N$ and N is endowed with a hyperbolic metric and $\mathcal{L}ag(M)$ consists of closed exact Lagrangians, as well as when $M = S^2$ and $\mathcal{L}ag(S^2)$ consists of equators.*

Corollary 1.3.0.2 is proved by studying the cone length of iterated distinguished triangles needed in order to approximate a Lagrangian as the approximation threshold ε goes to zero. Along the way, we define a persistence version of the notion of categorical entropy due to Dimitrov-Haiden-Katzarkov-Kontsevich [**DHKK14**], which we call *weighted categorical entropy*, and relate itemize to the barcode entropy introduced by Çineli-Ginzburg-Gurel [**cGG24**].

REMARK 1.3.0.3. The term "weighted" might be misleading. A candidate name for the above named refinement of categorical entropy would be *persistence* categorical entropy. However, we believe that persistence might not be a necessary tool in order to define approximability, and hence all the related notions (see Remark 1.1.0.5).

The rest of the chapter is dedicated to the proof of two geometric corollaries, that we now state.

COROLLARY 1.3.0.4 (Corollary 1.1.4 in [**ABC26**]). *Let $M = D^*N$ and assume that $\phi : D^*N \to D^*N$ is an exact symplectomorphism with support in the interior of D^*N. Let $\mathcal{F}_\varepsilon$ be a finite family of fibers of D^*N that is ε-approximating in the sense of Theorem 1.3.0.1 i. Let*

$$\chi(\phi; \mathcal{F}_\varepsilon) = \max_{F \in \mathcal{F}_\varepsilon} \left\{\max h_{\phi(F)} - \min h_{\phi(F)}\right\}$$

where $h_{\phi(F)}$ is the λ-primitive of the Lagrangian $F \in \mathcal{F}_\varepsilon$. Then, for any $L \in \mathcal{L}ag(M)$, we have

$$d_\gamma(L, \phi(L)) \leq 4(\varepsilon + N(L; \mathcal{F}_\varepsilon, \varepsilon)\chi(\phi; \mathcal{F}_\varepsilon)) \ .$$

In particular, if $\phi(F) = F$ for all $F \in \mathcal{F}_\varepsilon$, we have $d_\gamma(L, \phi(L)) \leq 4\varepsilon$, for every closed and exact Lagrangian L in D^*N.

For the next result, we recall the following notion introduced in [**BC07, BC09**]. Let L be a Lagrangian in a symplectic manifold M and $K \subset M$ be a subset. The relative Gromov width of L to K is defined as

$$\begin{aligned} \mathcal{W}(L; K) = \sup\Big\{\pi\frac{r^2}{2} \mid \exists\, j : (B_r, \omega_0) \to (M, \omega),\ j \text{ is an embedding}, \\ j^*\omega = \omega_0,\ \ j^{-1}(L) = B_r \cap \mathbb{R}^n,\ j(B_r) \cap K = \emptyset\Big\} \end{aligned}$$

where $(B_r, \omega_0) \subset (\mathbb{C}^n, \omega_0)$ is the standard symplectic r-ball in $\mathbb{C}^n$, centered at the origin.

COROLLARY 1.3.0.5. *Let M be one of the symplectic manifolds appearing in Theorem 1.3.0.1 and consider the associated class $\mathcal{L}ag^*(M)$ of Lagragians, endowed with the metric d_γ. For each $L \in \mathcal{L}ag^*(M)$ we have*

$$\mathcal{W}(L; \cup_{F\in\mathcal{F}_\varepsilon} F) \leq 2\varepsilon \ .$$

REMARK 1.3.0.6. The version we will prove in Chapter **??** (i.e. Corollary **??**, which is Corollary 1.1.5 in [**ABC26**]) is actually a more general version of Corollary 1.3.0.5.

1.4. Organisation of the thesis

Chapter 1: Algebraic preliminaries. The Fukaya categories appearing in this thesis will be filtered A_∞-categories over either the field $\mathbb{Z}_2$ or over the Novikov field over $\mathbb{Z}_2$. In §2.2 we define the version of the Novikov ring used in this thesis.

In §2.3 we introduce *dg*-categories and filtered *dg*-categories as categories enriched over chain complexes and filtered chain complexes. The monoidal symmetric category of chain complex is introduced in §2.3.1.

In §2.4 we introduce all the machinery around the concept of filtered A_∞-category used in this thesis.

In §**??** we introduce persistence categories [**BCZ24**] as categories enriched over the category of persistence modules and highlight a few properties. The symmetric monoidal category of persistence modules is introduced in §**??**.

In §**??** we briefly recall the definition of triangulated persistence category from [**BCZ24**] and its basic properties. Along the way, we recall the classical concept of triangulated category in §**??**.

In §**??** we first recall a few variants of a notion of approximability due to Turing [**Tur38**] and then introduce the concept of categorical metric approximability [**ABC26**] of a pseudo-metric space in a TPC and suggest a way to extend this concept.

The last section, §**??**, is dedicated to the (highly technical) concept of A_∞/persistence categories with increasing accuracy, which is needed to deal with filtered Fukaya categories. Along the way, we prove that the Hochschild homology of a system of A_∞-categories and the K-group of a system of TPCs are well-defined.

Chapter 2: Filtered Fukaya categories. In §**??** we work out the 'cluster' model for Fukaya categories and in §**??** we adapt the work of [**BCS21**] to show that our Fukaya categories naturally inherit the structure of weakly-filtered A_∞-categories with filtered unit.

In §**??** we construct the classes of ε-perturbation data, which ensure that the associated Fukaya categories are genuinely filtered. This proves a first part of Theorem 1.2.0.1, namely we shows that the space $\mathcal{P}$ is indeed non-empty. In §**??** we argue why the cluster model seems necessary to have a filtered Fukaya category.

In §**??** we develop the machinery of continuation functors between filtered Fukaya categories, and show that they are A_∞-functors with linear deviation, with deviation controlled by the size of domain and target.

We conclude this chapter by summarizing our constructions as a system of categories with increasing accuracy in **??**.

Chapter 3: Approximability for Lagrangian submanifolds. In this chapter we prove Theorem 1.3.0.1 and the corollaries mentioned in §1.3.

In §**??** we prove approximability of the space of closed and exact Lagrangians in disk cotangent bundles, that is, point *i.* in Theorem 1.3.0.1.

In §**??** we develop a retract-approximability criterion by studying open-closed maps in the persistence setting. We then use it to show (by explicit calculations) approximability of equators on S^2 and of non-contractible circles on $\mathbb{T}^2$, that is, points *ii.* and *iii.* in Theorem 1.3.0.1.

The last section, §**??**, is focused on weighted numerical complexity measurements for Lagrangian submanifolds and their applications. In particular, in this section are included the arguments necessary to deduce Corollary 1.3.0.2. The section also contains the proofs of Corollaries 1.3.0.4 and 1.3.0.5.

CHAPTER 2

Algebraic preliminaries

This chapter should be thought as a companion for the geometric applications in Chapter **??** and **??**. The definitions and results appearing in this chapter are mostly due to other authors. We decided to introduce the theory of *dg*-categories and persistence categories through the lenses of enriched category theory, which the author learned in [**Kel05, FS19**]. §2.3.2 is a rewriting of the basic standard theory of *dg*-categories (that the author learned in [**Kel06, Sei08**]) while §**??** and §**??** contains an introduction to the theory of persistence categories (originally introduced in [**BCZ24**]).

§2.4 contains an introduction to filtered A_∞-categories. Most of the material presented there is by now standard, with the exception of the discussion of of *dg*-functors (appearing in §2.3.4), filtered A_∞-functors (appearing in §**??**) and persistence Hochschild homology (appearing in §**??**). These results are due to the author and collaborators and were originally established in [**ABC26**].

§**??** contains a quick introduction to the theory of triangulated persistence categories (TPCs), as originally introduced in [**BCZ24**]. In §**??** we define the concept of "categorical metric approximability" for objects of TPCs. This concept was introduced in [**ABC26**] and is a quantitative generalization of the concept of generation and split-generation in triangulated categories, and can be seen - in a sense to be explained - as a genralization of the concept of total boundedness in the theory of metric spaces.

Finally, in §**??** we describe the notion of a *system of categories with increasing accuracy*, as defined in [**ABC26**]. Although technically involved, this framework is currently necessary for the study of persistence Fukaya categories.

2.1. Introduction

A_∞-categories were introduced by Fukaya in [**Fuk93**] as a homotopical generalization of *dg*-categories. In turn, *dg*-categories arose as a response to foundational difficulties in the theory of triangulated categories (see, for example, the survey [**Kel06**]). Triangulated categories, originally introduced by Verdier to formalize derived categories, appear naturally in many areas of mathematics. However, it is well known that they exhibit several undesirable categorical properties.

For instance, the bicategory of triangulated categories does not admit a monoidal structure. This deficiency reflects a more fundamental issue: mapping cones are not functorial. Concretely, given a triangulated category $\mathcal{T}$, there is no cone functor from the arrow category of $\mathcal{T}$ to $\mathcal{T}$.

The introduction of dg-categories was motivated by the desire to remedy these shortcomings. While this approach is successful from several perspectives (again, see [**Kel06**]), it comes at the cost of increased complexity. dg-categories encode a substantial amount of information, and A_∞-categories even more so. Consequently, when working with a (pre-triangulated) A_∞-category, one often ultimately discards part of the structure and passes to the associated triangulated homotopy category.

Filtered dg-categories and filtered A_∞-categories are natural refinements of their unfiltered counterparts, in which the algebraic structures are required to be compatible with a filtration on the morphism complexes indexed by the real line. In symplectic topology, the motivation for such structures is clear: Floer complexes are naturally filtered chain complexes whenever the action functional is defined (see Chapter **??**).

Just as the homology of a filtered chain complex naturally carries the structure of a persistence module, the homotopy category of a filtered dg- or A_∞-category inherits the structure of a "persistence category", namely a category enriched over the category of persistence modules.

2.2. The Novikov ring

In this section we introduce the Novikov ring and field over the field $\mathbb{Z}_2$ as a completion of rings of polynomials.

Let $G < \mathbb{R}$ be a subgroup of the additive real line. We recall the definition of the Novikov ring Λ_0^G associated to G over the field $\mathbb{Z}_2$. Let $G_{\geq 0} := G \cap \mathbb{R}_{\geq 0}$ and consider the polynomial ring $\mathbb{Z}_2[G_{\geq 0}]$; an arbitrary element of $\mathbb{Z}_2[G_{\geq 0}]$ will be denoted by $\sum_{i=0}^n a_i T^{\lambda_i}$, where $a_i \in \mathbb{Z}_2$ and $\lambda_i \in G_{\geq 0}$ are ordered increasingly, i.e. $\lambda_0 \leq \lambda_1 \leq \cdots \leq \lambda_n$. For any $r \in \mathbb{R}_{\geq 0}$ we define the ideal

$$I(r) := \left\{ \sum_{i=0}^n a_i T^{\lambda_i} \in \mathbb{Z}_2[G_{\geq 0}] : \; r < \lambda_0 \in G_{\geq 0} \right\}.$$

We then define the Novikov ring associated to G over $\mathbb{Z}_2$ as

$$\Lambda_0^G := \varprojlim \frac{\mathbb{Z}_2[G_{\geq 0}]}{I(r)}$$

and denote its field of fractions by Λ^G, which is called the Novikov field associated to G. In the case $G = \mathbb{R}$, $\Lambda_0 := \Lambda_0^{\mathbb{R}}$ will be called the (universal) Novikov ring, and $\Lambda := \Lambda^{\mathbb{R}}$ the (universal) Novikov field.

We will usually write

$$\Lambda := \left\{ \sum_{i=0}^\infty a_i T^{\lambda_i} : \; a_i \in \mathbb{Z}_2, \; \lambda_i \in \mathbb{R} \text{ such that } \lim_{i \to +\infty} \lambda_i = +\infty \right\},$$

and

$$\Lambda_0 := \left\{ \sum_{i=0}^\infty a_i T^{\lambda_i} \in \Lambda : \; \lambda_i \in [0, +\infty) \right\}.$$

Given an element $P(T) = \sum_{i=0}^{\infty} a_i T^{\lambda_i} \in \Lambda$ (or Λ_0), we will always assume that $a_0 = 1$ and $\lambda_0 \leq \lambda_i$ for any $i \geq 1$.

2.3. Filtered dg-categories as enriched categories

In this section we introduce dg-categories as categories enriched over chain complexes and filtered dg-categories as categories enriched over filtered chain complexes. Most of the content of this section is well-known to experts. We included this material, which do not appear neither in [**Amb25**] nor in [**ABC26**], in the thesis as we believe that (i) it provides a motivation behind some of the contructions developed in [**ABC26**] (and outlined later in this chapter) with filtered A_∞-categories and (ii) we believe that the language of enriched categories can be used in the case of A_∞-and triangulated categories in order to better understand the concept of approximability (outlined in §**??**). The latter point will not be undertaken in this thesis but will be the subject of future work.

2.3.1. Categories of chain complexes. Let R be a commutative ring with unit. For us, a chain complex over R will be a graded chain complex of R-modules, that is a graded R-module $C = \oplus_{n\in\mathbb{Z}} C_n$ together with moprhisms of modules $d_n \colon C_n \to C_{n-1}$ such that $d_n \circ d_{n+1} = 0$. As usual in the literature, we will often drop the degree $n \in \mathbb{Z}$ from the notation.

We denote by $\mathbf{Ch}(R)$ the following category of chain complexes of chain complexes over R. Objects of $\mathbf{Ch}(R)$ are chain complexes over R and morphisms are chain maps between them. We denote by $\mathbf{Ch}^{\mathrm{fg}}(R)$ the full subcategory of chain complexes over finitely generated R-modules, which from now on will be called finitely generated chain complexes over R; $\mathbf{Ch}^{\mathrm{b}}(R)$ the full subcategory of bounded chain complexes over R; $\mathbf{Ch}^{\mathrm{proj}}(R)$ the full subcategory of chain complexes over projective R-modules, which from now on will be called projective chain complexes over R. We allow for obvious mixes in the notation: for instance, $\mathbf{Ch}^{\mathrm{b,fg,proj}}(R)$ is the category of bounded finitely generated projective chain complexes over R.

One remarkable property of $\mathbf{Ch}(R)$ is that it comes with a canonical notion of cone.

DEFINITION 2.3.1.1. Let (C, d_C) and (D, d_D) be chain complexes over R and $f \colon C \to D$ a chain map. The cone of f is the chain complex

$$\mathrm{Cone}(f) := D \oplus C[1]$$

with differential

$$d_{\mathrm{Cone}} := \begin{pmatrix} d_D & f \\ 0 & d_{C[1]} \end{pmatrix}$$

where $C[1]$ is the chain complex C shifted down by 1 in degree.

Let f be as above, then the sequence of chain complexes

$$0 \to C \xrightarrow{f} D \to \mathrm{Cone}(f) \to 0$$

where the third map is the inclusion into the first factor, is exact. We call this sequence the cone sequence associated to f.

Let (C, d_C) and (D, d_D) be chain complexes. We define their tensor product $(C, d_C) \otimes (D, d_C)$ as the chain complex defined on degree $n \in \mathbb{Z}$ as the R-module $\bigoplus_{p+q=n} C_p \otimes_R D_q$ with differential

$$d_\otimes := d_C \otimes id_D + id_C \otimes d_D.$$

The following result is an exercise.

LEMMA 2.3.1.2. *The category* $\mathbf{Ch}(R)$ *is a closed symmetric monoidal category when endowed with the tensor* $\otimes$ *as monoidal product and the trivial chain complex* $(R, 0)$ *as unit.*

It is straightforward to see that both the cone and tensor product constructions are well-defined in all of the subcategories of $\mathbf{Ch}(R)$ introduced at the beginning of this section. Note that $\mathbf{Ch}(R)$ is not compact closed, while $\mathbf{Ch}^{\mathrm{b,fg,proj}}$ is.

We recall that closedness means that the right tensor functor $- \otimes C$ has a right adjoint for any chain complex C over R. Let $C_0 = (C_0, d_0), C_1 = (C_1, d_1)$ and $C_2 = (C_2, d_2)$ be chain complexes over R. Dropping differential from the notation, Closedness of $\mathbf{Ch}(R)$ means that for any three chain complexes $(C_0, d_0), (C_1, d_1)$ and (C_2, d_2) there is a chain complex $[C_1, C_2]$ such that

$$\hom_{\mathbf{Ch}(R)}(C_0 \otimes C_1, C_2) \cong \hom_{\mathbf{Ch}(R)}(C_0, [C_1, C_2]).$$

It is an exercise to realise that $[C_1, C_2]$ is the complex of graded linear maps $f \colon C_1 \to C_2$ endowed with the differential

$$\partial f := d_2 \circ f + f \circ d_1. \tag{1}$$

Suppose we are given a chain map $g \colon C_0 \otimes C_1 \to C_2$. This means that for any $c_0 \otimes c_1 \in C_0 \otimes C_1$ we have

$$g(d_0 c_0 \otimes c_1) + g(c_0 \otimes d_1 c_1) + d_2 g(c_0 \otimes c_1) = 0$$

By writing $g_{c_0}(c_1) := g(c_0 \otimes c_1)$ this can be rewritten as

$$g_{dc_0}(c_1) + (d_2 g_{c_0}(c_1) + g_{c_0}(d_1 c_1)) = 0.$$

This provides an hint on how g can be seen as a chain map in $\hom_{\mathbf{Ch}(R)}(C_0, [C_1, C_2])$; the shift in grading depends on the input c_0 and appears as a result of the definition of $\otimes$ on $\mathbf{Ch}(R)$.

REMARK 2.3.1.3. Indeed, we observe that a generalized element of $[B, C]_R$, i.e. a chain map $R \to [B, C]_R$ is precisely a choice of chain map $B \to C$.

The complex $[C_1, C_2]$ is called the complex of internal morphisms. It is a standard result that the category with the same objects as $\mathbf{Ch}(R)$ and morphism spaces given by complexes of internal morphisms is a category enriched over $\mathbf{Ch}(R)$. In the case of chain complexes, the resulting composition is the standard composition of graded maps. We denote this category by $\mathbf{Ch}_{dg}(R)$ and call it the dg-category of chain complexes. The following is a consequence of standard abstract non-sense.

LEMMA 2.3.1.4. *The category $\mathbf{Ch}_{dg}(R)$ is a closed symmetric monoidal category when endowed with the tensor $\otimes$ as monoidal product and the trivial chain complex $(R, 0)$ as unit.*

We can define yet another category of chain complexes, which will be denoted by $H_0(\mathbf{Ch}_{dg}(R)$ and called the homological category of $\mathbf{Ch}_{dg}(R)$. Its objects are chain complexes over R and its morphism spaces are the homologies of the morphism spaces of $\mathbf{Ch}_{dg}(R)$. It is well-known that he homological category $H^0(\mathbf{Ch}_{dg}(R))$ of $\mathbf{Ch}_{dg}(R)$ (and of all the subcategories introduced above) is a triangulated category, with exact triangles induced by sequences isomorphic to a cone sequence. It is easy to see that this category is isomorphic to the homotopy category of chain complexes, that is the category $\mathbf{Ch}(R)$ with morphism spaces quotiented by chain homotopies.

2.3.2. *dg*-categories. The discussion in the previous section motivates the following definition.

DEFINITION 2.3.2.1. A dg-category over R is a category enriched over $\mathbf{Ch}(R)$.

Unpacked, what the above definition means is that a dg-category A is a category where the morphisms space $\hom_A(L_0, L_1)$ is a chain complex for any couple of objects L_0 and L_1 of A, composition is a chain map – which, due to the definition of the tensor product above, means that the composition is compatible with differentials via the Leibniz rule – and for all objects L of A there is an element $e_L \in \hom_A(L, L)$ which is a cycle and is a multiplicative identity for the composition. The element e_L will be called the unit of L.

The following is immediate.

LEMMA 2.3.2.2. *The category $\mathbf{Ch}_{dg}(R)$ is a dg-category.*

We denote by $dg(R)$ the category of dg-categories over R. It naturally admits the structure of a bicategory.

DEFINITION 2.3.2.3. Let A and B be dg-categories. An enriched functor $F\colon A \to B$ is called a dg-functor. An enriced natural transformation $\eta\colon F \to G$ between two dg-functors $F, G\colon A \to B$ is called a dg-natural transformation.

Unpacked this means that a dg-functor $F\colon A \to B$ is a functor such that for any objects L_0 and L_1 of A, the map

$$F\colon \hom_A(L_0, L_1) \to \hom_B(FL_0, FL_1)$$

is a chain map. A dg-natural transformation $\eta\colon F \to G$ is a natural transformation such that for any object L of A the element $\eta_L \in \hom_B(FL, GL)$ is of degree zero[1] and closed (i.e. a cycle with respect to the differential on the chain complex $\hom_B(FL, GL)$).

As hinted aove, $dg(R)$ is a bicategory, that is in particular, for any dg-categories A and B over R, dg-functors $A \to B$ together with dg-natural transformations form a category $\hom_{dg(R)}(A, B)$.

The following is a standard consequence of Lemma 2.3.1.2.

[1]We recall that a natural transformation $\eta\colon F \to G$ is graded of degree d if for any object L of A the morphism η_L in the chain complex $\hom_B(FL, GL)$ is of degree d.

LEMMA 2.3.2.4. *The category $dg(R)$ is closed symmetric monoidal, with monoidal functor $\otimes_{dg}$ induced by the monoidal structure of $\mathbf{Ch}_{dg}(R)$ and unit given by the ring R seen as a dg-category with a single object and trivial differential.*
Given two dg-categories B and C, the dg-category of internal morphisms $[B, C]_{dg}$ appearing in the isomorphisms

$$\hom_{dg(R)}(A \otimes B, C) \cong hom_{dg(R)}(A, [B, C]_{dg})$$

has the same objects of $\hom_{dg}(B, C)$ (that is, dg-functors $B \to C$). Let $F, G \colon B \to C$ be dg-functors, then the morphism complex $\hom_{[B,C]_{dg}}(F, G)$ consists of graded natural transformations $\eta \colon F \to G$ with differential induced object-wise by the differential on morphism spaces in the dg-category C. In other words, given a natural transformation $\eta \colon F \to G$ and an object L of B we have

$$(d\eta)_L = d(\eta_L)$$

where the first differential is on $\hom_{[B,C}(F, G)$ and the second on $\hom_C(FL, GL)$. Composition of morphisms in $[B, C]$ is the usual composition of graded natural transformations.

We omit the proof of this Lemma.

REMARK 2.3.2.5. Let's stop for a moment to observe the following seemingly surprising fact. In the case of chain complexes, internal morphisms consist of graded linear maps (not necessarily chain maps), while in the case of dg-categories, objects of the internal morphism dg-category are still dg-functors, and not something more general (as, for instance "graded linear functors"). The difference lies at the level of natural transformations. One can convince themselves that this is indeed correct by considering morphisms between tensors of morphisms between tensor dg-categories: the shift in grading "absorbed" by the fact that the target is a tensor of chain complexes, in contrast with the case of $\mathbf{Ch}(R)$.

The following definition is, in the case of dg-categories, redundant, but we hope it provides some insight in the A_∞-case (cfr. [**Sei08**, Section (1d)]).

DEFINITION 2.3.2.6. Let B and C be dg-categories over R. Then a morphism in the dg-category $[B, C]_{dg}$ is called a pre-dg-natural transformation.

Let A be a dg-category. We recall, that this means that morphisms spaces in A are chain complexes. We can then define a category by taking the homology of these complexes. In other words, we define a category $H_0(A)$ with the same objects as A and, given objects L_0 and L_1 of A, morphisms

$$\hom_{H_0(A)}(L_0, L_1) := H_0(\hom_A(L_0, L_1))$$

where H_0 is the homology functor in degree 0. We call this the homotopy category of A. This is a preadditive category, which is moreover enriched over $R-\mathrm{mod}$. Similarly, we can define the category enriched over graded R-modules $H_*(A)$ by considering the homology of morphism spaces in all degrees, this will be called the graded homotopy category of A.

We briefly elaborate on a point that is often source of confusion. Being a bicategory, $dg(R)$ comes with a notion of equivalence, that is an adjunction where the natural transformations are isomorphisms. Equivalently, a dg-functor $F\colon A \to B$ is said to be an enriched equivalence (from now, a dg-equivalence) if and only if F is fully faithful and dense on objects. What this means, is that on morphisms

$$F\colon\ \hom_A(L_0, L_1) \to \hom_B(FL_0, FL_1)$$

is an isomorphism in $\mathbf{Ch}_R$, i.e. a chain isomorphism. This notion is often too restrictive in the case of concrete applications (see, for instance, §**??**). This is why the weaker notion of quasi-equivalence is often preferred.

Definition 2.3.2.7. A dg-functor $F\colon A \to B$ is said to be a quasi-equivalence if and only if it is dense on objects and the chain maps induced on morphism spaces are quasi-isomorphisms of chain complexes.

Remark 2.3.2.8. At this point we want to highlight how A_∞-structure naturally arise from an homotopical treatment of dg-categories. It was discovered by Tabauda in [**Tab05**] that the category $dg(R)$ admits the structure of a Quillen model category, with weak equivalences given by quasi-equivalences and fibrations given by certain degreewise surjecive dg-functors. It was proved in [**Toë07**] that the homotopy category $\mathrm{Ho}(dg(R))$ of $dg(R)$ is a closed symmetric monoidal category, and Kontsevich proved that the dg-category $[A, B]_{\mathrm{Ho}(dg(R))}$ can be identified with the dg-category of A_∞-*functors* from A to B. We will introduce this dg-category, denoted $\hom_{A_\infty}(A, B)$ in §**??**.

2.3.3. The dg-category of dg-(bi)modules. In this subsection we define dg-modules and dg-bimodules. We fix a dg-category A over R.

Lemma 2.3.3.1. *For any object L of A, there is a dg-functor*

$$\mathcal{Y}^A_L\colon A \to \mathbf{Ch}_{dg}(R)$$

given by $\mathcal{Y}^A_L(L') := \hom_A(L', L)$ *on objects and, given a morphism* $\hom_A(L_0, L_1)$*, the image* $\mathcal{Y}^A_L(f) \in \hom_{\mathbf{Ch}_{dg}(R)}(\hom_A(L_0, L), \hom_A(L_0, L))$ *is given by postcomposition with f.*

Proof. The fact that the functor is well-defined at the level of objects follows from the definition of dg-category as a category enriched over $\mathbf{Ch}(R)$. In order for $\mathcal{Y}^A_L$ to be a dg-functor, we have to check that the induced map

$$\mathcal{Y}^A_L\colon\ \hom_A(L_0, L_1) \to \hom_{\mathbf{Ch}_{dg}(R)}(\hom_A(L_0, L), \hom_A(L_1, L)$$

is a morphism in

$$\hom_{\mathbf{Ch}(R)}\big(\hom_A(L_0, L_1), \hom_{\mathbf{Ch}_{dg}(R)}(\hom_A(L_0, L), \hom_A(L_1, L)\big)$$

i.e. a chain map.

We recall that a morphism in the category $\mathbf{Ch}_{dg}(R)$ is an internal hom in $\mathbf{Ch}(R)$, that is a graded linear map, and the differential is defined via Eq. (1). Let $f \in \hom_A(L_0, L_1)$. Then for $g \in \hom_A(L_0, L)$ we have

$$(\partial \mathcal{Y}_L^A)(f) = d(g \circ f) + (dg \circ f) = ((dg \circ f) + (g \circ df)) + (dg \circ f) = \mathcal{Y}_L^A(df)$$

and this proves the claim. □

REMARK 2.3.3.2. A dg-functor $A \to \mathbf{Ch}_{dg}(R)$ is also called a $\mathbf{Ch}(R)$-valued presheaf on A^{op}. If such a functor isomorphic (in the category of dg-categories) to a functor of the form $\mathcal{Y}_L^A$ for some L, it is called a representable sheaf.

We recall that the dg-category of dg-functors between dg-categories B and C has been defined as the dg-category $[B, C]_{dg(R)}$ of internal homs in the category of dg-categories $dg(R)$.

LEMMA 2.3.3.3. *There is a dg-functor*

$$\mathcal{Y}^A \colon A \to [A, \mathbf{Ch}_{dg}(R)]$$

given by $\mathcal{Y}^A(L) := \mathcal{Y}_L^A$ *on objects and, given a morphism* $f \in \hom_A(L_0, L_1)$*, the image*

$$\mathcal{Y}^A(f) \in \hom_{[A, \mathbf{Ch}_{dg}(R)]}(\mathcal{Y}_{L_0}^A, \mathcal{Y}_{L_1}^A))$$

is given by postcomposition with f*.*

PROOF. The fact that, given an object L of A, the image $\mathcal{Y}^A(L) = \mathcal{Y}_L^A$ is a dg-functor is proved in Lemma 2.3.3.1. The fact that $\mathcal{Y}^A(f)$ is a natural transformation is immediate. □

The following is a consequence of the strong Yoneda Lemma in enriched category theory.

THEOREM 2.3.3.4 ((left) Yoneda lemma for dg-categories). *Let L be an object of A. There is an isomorphism*

$$\hom_A(L_0, L_1) \to \hom_{[A, \mathbf{Ch}_{dg}(R)]}(\mathcal{Y}_{L_0}^A, \mathcal{Y}_{L_1}^A))$$

in $\mathbf{Ch}(R)$*.*

The functor $\mathcal{Y}^A$ will be called the Yoneda embedding functor of A.

DEFINITION 2.3.3.5. A left dg-module on A is a dg-functor $A \to \mathbf{Ch}_{dg}(R)$. A right dg-module over A is a dg-functor $A^{op} \to \mathbf{Ch}_{dg}(R)$. If B is another dg-category, a dg-bimodule over (A, B) is a dg-functor $A^{op} \otimes_{dg} B \to \mathbf{Ch}_{dg}(R)$.

We briefly unpack the definition of left dg-module. Let $M \colon A \to \mathbf{Ch}_{dg}(R)$ be a dg-functor, then for any object L of A we have a chain complex $M(L)$. Let $f \in \hom_A(L_0, L_1)$ be a morphism in A, then $M(f)$ will be an element of $\hom_{\mathbf{Ch}_{dg}(R)}(M(L_0), M(L_1))$. We will not look at morphisms this way, but rather as follows. Given L_0 and L_1 objects of A, M prescibes a morphism in

$$\hom_{\mathbf{Ch}(R)}(\hom_A(L_0, L_1), \hom_{\mathbf{Ch}_{dg}(R)}(M(L_0), M(L_1))).$$

Since $\hom_{\mathbf{Ch}_{dg}(R)}(M(L_0), M(L_1)) := [M(L_0), M(L_1)]_{\mathbf{Ch}(R)}$, we get that M indeed prescribes a chain map

$$\hom_A(L_0, L_1) \otimes M(L_0) \to M(L_1)$$

for any object L_0 and L_1. We will denote this map by $\circ_M$.
Similarly, a right dg-module N over A consists of a collection $N(L)$ of chain complexes, one for every object L of A, together with chain maps $\hom_A(L_0, L_1) \otimes N(L_1) \to N(L_0)$.
A dg-bimodule P over (A, B) consists of a collection $P(L, K)$ of chain complexes, for any object L of A and K of B, together with chain maps

$$\hom_A(L_0, L_1) \otimes \hom_B(K_0, K_1) \otimes P(L_1, K_0) \to P(L_0, K_1).$$

Equivalently, we can see P as a collection $P(L, K)$ as above together with chain maps

$$\hom(L_0, L_1) \otimes P(L_1, K) \to P(L_0, K) \text{ and } P(L, K_0) \otimes \hom_B(K_0, K_1) \to P(L, K_1)$$

satisying the obvious associativity considitions.

We write

$$\mathrm{mod}^l_A := [A, \mathbf{Ch}_{dg}(R)]_{dg(R)}, \qquad \mathrm{mod}^r_A := [A^{op}, \mathbf{Ch}_{dg}(R)]_{dg(R)}$$

and

$$\text{and } \mathrm{bimod}_{(A,B)} := [A^{op} \otimes_{dg} B, \mathbf{Ch}_{dg}(R)]_{dg(R)}.$$

It is clear that these are dg-categories. We will call a natural transformation between dg-(bi)modules a pre-morphism of dg-(bi)modules, while dg-natural transformations (i.e. cycles) will be called morphisms of dg-(bi)modules.

Given a dg-cateogory A we write $\mathcal{Y}^A(A) \subset \mathrm{mod}^l_A$ for the image of the Yoneda embedding of A. This is a full-subcategory of mod^l_A, and its elements are called Yoneda (or representable) dg-modules. We define $\overline{\mathcal{Y}^A(A)} \subset \mathrm{mod}^l_A$ to be the closure of $\mathcal{Y}^A(A)$ with respect to quasi-isomorphisms. This is a full-subcategory of mod^l_A, and we call its objects quasi-representable dg-modules. Of course, $\mathcal{Y}^A(A)$ and $\overline{\mathcal{Y}^A(A)}$ are quasi-equivalent dg-categories.

2.3.4. Pullback and pushforward(s) of dg-modules. In this section we recall the definition of the pullback of a dg-functor and develop a model for a notion of pushforward. The goal of this section is to motivate the definition of the filtered pushforward of A_∞-functors with linear deviation in §**??**.

Let $F\colon A \to B$ be a dg-functor. For the moment we restrict ourselves to left dg-modules. There is a canonical dg-functor $F^*\colon \mathrm{mod}^l_B \to \mathrm{mod}^l_A$ induced by precomposition with F (see for instance [**Sei08**]). We call F^* the pullback of F to dg-modules. In the above interpretation of left dg-modules as collection of chain complexes with decoration, the action of F^* can be described as follows. Let N be a dg-module over B, then F^*N is the dg-module over A such that

$$F^*N(L) := N(FL)$$

for any object L of A and comes with the chain map

$$\circ_{F^*N} := - \circ_N F(-)\colon \hom_A(L_0, L_1) \otimes F^*N(L_0) \to F^*N(L_1).$$

Let $f\colon N_0 \to N_1$ be a pre-morphism of dg-modules over B, then we set $F^*f\colon F^*N_0 \to F^*N_1$ to be the pre-morphism of dg-modules over A given as $F^*f(x) := f(x)$ for any $x \in \hom_A(F^*N_0L, F^*N_1L)$.

We would like to define a functor $\mathrm{mod}^l_A \to \mathrm{mod}^l_B$ adjoint to F^*. In other words, we want to construct left and right Kan extensions for F together with any dg-module M over A.

We recall that, given a left dg-module N over B, the object $\hom_{\mathrm{mod}^l_B}(\hom_B(-, F(-)), N)$ can be seen as a left dg-module over A, with multiplication

$$\hom_A(L_0, L_1) \otimes \hom_{\mathrm{mod}^l_B}(\hom_B(-, F(L_1)), N) \to \hom_{\mathrm{mod}^l_B}(\hom_B(-, F(L_0)), N)$$

given by sending a pure tensor $a \otimes f$ to the pre-morphism af of left dg-modules over B given by

$$(af)(b) := f(F(a) \circ b)$$

for $b \in \hom_B(K, F(L_0))$.

The following is a consequence of the Yoneda lemma.

Lemma 2.3.4.1. *The functor $F^*\colon mod^l_B \to mod^l_A$ is quasi-isomorphic to the functor*

$$\hom_{mod^l_B}(\mathcal{Y}^B \circ F, -)\colon mod^l_B \to mod^l_A.$$

Note the following. The functor $\mathcal{Y}^B \circ F\colon A \to \mathrm{mod}^l_B$ can be seen as a (B, A)-bimodule $\underline{F}$ with the obvious composition maps. It is well-known that a left adjoint of the functor

$$\hom_{\mathrm{mod}^l_B}(\underline{F}, -)\colon \mathrm{mod}^l_B \to \mathrm{mod}^l_A$$

is the functor $\mathrm{mod}^l_A \to \mathrm{mod}^l_B$ given by the convolution tensor product $\underline{F} \otimes_A -$ that we now define. Let M be a dg-module over A, then $\underline{F} \otimes_A M$ is the dg-module over B defined via

$$(\underline{F} \otimes_A M)(K) := \bigotimes_{L \in \mathrm{Obj}(A)} \underline{F}(K, L) \otimes_{\mathbf{Ch}(R)} M(L)$$

on an object K of B, with the obvious differentials coming from the dg-structure of B and the dg-module structure of M, and composition maps coming from the dg-bimodule structure of $\underline{F}$. Given a morphism $f \in \hom_{\mathrm{mod}^l_A}(M_0, M_1)$ of dg-modules, we define $\underline{F} \otimes_A f \in \hom_{\mathrm{mod}^l_F}(\underline{F} \otimes_A M_0, \underline{F} \otimes_A M_1)$ via

$$\underline{F} \otimes_A f(b \otimes m) := b \otimes f(m).$$

Corollary 2.3.4.2. *We have $\underline{F} \otimes_A - \dashv F^*$.*

Example 2.3.4.3. The fundamental problem of $\otimes_A$ is that it does not preserve quasi-isomorphisms. A very simple example goes as follows. Let $R = \mathbb{Z}$ and consider $A = B$ to be the dg-category with one object. A dg-module over A is a chain complex of abelian groups. Let M be the chain complex $\mathbb{Z}_2$ concentrated in degree 0 and N be the chain complex

$$0 \to \mathbb{Z} \to \mathbb{Z} \to 0$$

where the middle map is multiplication by 2, the first $\mathbb{Z}$ lies in degree 1 and the second in degree 0. Then N is quasi-isomorphic to M. It is easy to see that if we consider $\mathbb{Z}_2$ as a dg-bimodule we have $\mathbb{Z}_2 \otimes_{\mathbb{Z}} M \cong M$ while $\mathbb{Z}_2 \otimes_{\mathbb{Z}} N$ is the complex $0 \to \mathbb{Z}_2 \to \mathbb{Z}_2 \to 0$, which has non-trivial first homology.

The issue raised in Example 2.3.4.3 is solved by using the machinery of Quillen adjunction. In particular, since the pullback is well-behaved with respect to the Quillen model structure on dg-categories of dg-modules (see [**Toe11**]), it induces an adjunction at the level of derived categories. In particular, the functor $\underline{F} \otimes_A -$ induces a derived functor $\underline{F} \otimes^{\mathbb{L}}_A -$ by replacing a dg-module with a cofibrant replacement. Since in the canonical Quillen model structure on the category mod^l_A every object is fibrant, this functor is adjoint to F^* seen as a functor on derived categories.

Example 2.3.4.4. Note that N above is a free resolution of M, in particular projective, i.e. a cofibrant replacement in the standard projective model category structure on bounded chain complexes. In this case, it is immediate that to see that the derived tensor product preserves the quasi-isomorphism.

In the standard model structure on mod^l_A, there is a standard cofibrant replacement to any dg-module M via the so-called bar construction. Let M be a dg-module over A. Then we define $\overline{M}$ to be the dg-module defined via

$$\overline{M}(L) := \bigoplus_{d \geq 0} \bigoplus_{L_0, \ldots, L_d \in \mathrm{Obj}(A)} A(\vec{L}) \otimes M(L_d)$$

with the obvious decorations, where we used the notation $\vec{L} := (L_0, \ldots L_d)$ and $A(\vec{L}) := \hom_A(L_0, L_1) \otimes \cdots \otimes \hom_A(L_{d-1}, L_d)$. It is well-known that $\overline{M}$ is cofibrant (since it is semi-free) and quasi-isomorphic to M (indeed, in the derived category $\overline{M} \cong \Delta_A \otimes^{\mathbb{L}}_A M \cong M$, where Δ_A is the diagonal bimodule of A).

The above discussion motivates the following definition.

Definition 2.3.4.5. Let $F\colon A \to B$ be a dg-functor. We define the pushforward functor induced by F on left dg-modules to be the functor

$$F_*\colon \mathrm{mod}^l_A \to \mathrm{mod}^l_B$$

defined as follows:

(1) Given a left dg-module M over A and an object K of B, define $F_*M(K)$ to be the chain complex

$$F_*M(K) := \bigoplus_{d\geq 0} \bigoplus_{L_0,\dots,L_d \in \mathrm{Obj}(A)} \underline{F}(K, L_0) \otimes A(\vec{L}) \otimes M(L_d)$$

with differential induced by the differentials on A, B and M as well as composition on A, left-module structure on M and right-module structure on F; the left-composition for F_*M is induced by the left-composition of F.

(2) given a pre-morphism $f\colon M_0 \to M_1$ of dg-modules over A, i.e. a natural transformation, we define the pre-morphism $F_*f\colon F_*M_0 \to F_*M_1$ via

$$F_*f(b \otimes a_1 \otimes \cdots \otimes a_d \otimes m) := b \otimes a_1 \otimes \cdots \otimes a_d \otimes f(m)$$

for a pure tensor in F_*M_0 applied to some object of B, and extend linearly.

The following lemma is a simple exercise.

LEMMA 2.3.4.6. *Let L be an object of A, then $F_*\mathcal{Y}^A_L$ is quasi-isomorphic to $\mathcal{Y}^B_{FL}$. In particular, F_* preserves the class of quasi-representable modules.*

REMARK 2.3.4.7. We are not aware of how to define a right-adjoint to F^*. All the naive attempts via manipulation of bimodules induced by functors seem to send left modules to right ones. Hence, additional assumptions such as the Calabi-Yau property (see [**Gan12**]) seem necessary. Anyway, in the followings we are gonna deal with pushforwards of quasi-equivalences only. In this case, we believe that, when defined, the right adjoint to the pullback is quasi-isomorphic to the left one described above.

2.3.5. The category of filtered chain complexes. In our geometric examples, we will deal with chain complexes defined over fields that carry non-trivial persistence properties. Hence, a bit of care is required when talking about filtered chain complexes.

Let $\mathbf{k}$ be a field and $\mathbf{k}_0$ be a subring of $\mathbf{k}$. Assume there is a filtration $(\mathbf{k}^\alpha)_{\alpha\in\mathbb{R}}$ of $\mathbf{k}$ by $\mathbf{k}_0$-modules such that $\mathbf{k}^0 = \mathbf{k}_0$ and for all $\alpha, \beta \in \mathbb{R}$ we have

$$\mathbf{k}^\alpha \otimes_{\mathbf{k}_0} \mathbf{k}^\beta \subset \mathbf{k}^{\alpha+\beta}.$$

Note that, since $\mathbf{k}_0$ is a subring of $\mathbf{k}$, it contains in particular the multiplicative identity $1_{\mathbf{k}}$.

EXAMPLE 2.3.5.1. In this thesis, we will work with two instances of the above structure.

(1) An example is the Novikov field $\mathbf{k} = \Lambda$ over $\mathbb{Z}_2$ together with $\mathbf{k}_0 = \Lambda_0$ and

$$\mathbf{k}^\alpha := \Lambda^\alpha := T^{-\alpha}\Lambda_0.$$

(2) The second example is the field $\mathbf{k} = \mathbb{Z}_2$ with $\mathbf{k}_0 = \mathbb{Z}_2$ and the filtration

$$\mathbf{k}^\alpha := \begin{cases} \mathbb{Z}_2, & \text{if } \alpha \geq 0, \\ 0, & \text{if } \alpha < 0. \end{cases}$$

DEFINITION 2.3.5.2. A filtered chain complex over $(\mathbf{k}, \mathbf{k}_0)$ consists of a chain complex (C, d) over $\mathbf{k}$ together with an increasing filtration $(C^{\leq\alpha}_{\alpha\in\mathbb{R}}$ such that

$$d(C^{\leq\alpha}) \subset C^{\leq\alpha}$$

for all $\alpha \in \mathbb{R}$,

$$\mathbf{k}^{\alpha} \otimes_{\mathbf{k}_0} C^{\leq\beta} \subset C^{\leq\alpha+\beta}$$

for all $\alpha, \beta \in \mathbb{R}$ and moreover

$$\lim_{\alpha\to\infty} C^{\leq\alpha} = C.$$

EXAMPLE 2.3.5.3. An example of filtered chain complex over $(\mathbf{k}, \mathbf{k}_0)$ is given by $\mathbf{k}$ with trivial differential, together with $\mathbf{k}_0$ the filtration $()\mathbf{k}^{\alpha})_{\alpha}$. We denote this filtered chain complex by $\mathbf{k}$.

DEFINITION 2.3.5.4. Let (C, d_C) and (D, d_D) be filtered chain complexes over $(\mathbf{k}_0, \mathbf{k})$. A graded $\mathbf{k}$-linear map $f\colon C \to D$ is said to be filtered if it restricts to $\mathbf{k}_0$-linear maps

$$f^{\leq\alpha} := f|_{C^{\leq\alpha}}\colon C^{\leq\alpha} \to D^{\leq\alpha}$$

for any $\alpha \in \mathbb{R}$.

It is an exercise to see that filtered chain complexes over $(\mathbf{k}_0, \mathbf{k})$ together with filtered chain maps form a category, denoted by $\mathcal{F}\mathbf{Ch}(\mathbf{k}, \mathbf{k}_0)$.

Let (C, d_C) and (D, d_D) be filtered chain complexes over $(\mathbf{k}_0, \mathbf{k})$. We define the tensor product $C \otimes_{(\mathbf{k}_0,\mathbf{k})} D$ to be the filtered chain complex over $(\mathbf{k}_0, \mathbf{k})$ consisting of the chain complex $C \otimes_{\mathbf{k}} D$ over $\mathbf{k}$ together with the filtration $((C \otimes_{(\mathbf{k}_0,\mathbf{k})} D)^{\leq\alpha})_{\alpha\in\mathbb{R}}$ defined via

$$(C \otimes_{(\mathbf{k}_0,\mathbf{k})} D)^{\leq\alpha} := \underset{\alpha_1+\alpha_2\leq\alpha}{\underrightarrow{\operatorname{colim}}}\, C^{\leq\alpha_1} \otimes_{\mathbf{k}_0} D^{\leq\alpha_2}.$$

LEMMA 2.3.5.5. *The category $\mathcal{F}\mathbf{Ch}(\mathbf{k}, \mathbf{k}_0)$ of filtered chain complexes over $(\mathbf{k}, \mathbf{k}_0)$ is a closed symmetric monoidal category when endowed with the tensor product $\otimes$ as monoidal product and the filtered chain complex $\mathbf{k}$ as unit.*

DEFINITION 2.3.5.6. A filtered dg-category over $(\mathbf{k}, \mathbf{k}_0)$ is a category enriched over $\mathcal{F}\mathbf{Ch}(\mathbf{k}, \mathbf{k}_0)$.

We briefly unpack this definition. Let A be a filtered dg-category over $(\mathbf{k}, \mathbf{k}_0)$. Then for any two objects L_0 and L_1 of A, the morphism space $\hom_A(L_0, L_1)$ is a filtered chain complex. We will denote the filtration by $\hom_A^{\leq\alpha}(L_0, L_1)$. Then A can be seen as a dg-category where the composition is a filtered chain map, and such that for any object L of A the unit e_L lies in $\hom_A^{\leq 0}(L, L)$. The first property is immediate, while the second is equivalent to the choice of a filtered chain map $\mathbf{k} \to \hom_A(L, L)$, which is prescribed by the notion of enrichment.

The prototipical example of filtered dg-category over $(\mathbf{k}, \mathbf{k}_0)$ is the internal category of the closed monoidal category $\mathcal{F}\mathbf{Ch}(\mathbf{k}, \mathbf{k}_0)$, which we will denote by $\mathcal{F}\mathbf{Ch}_{\mathcal{F}dg}(\mathbf{k}, \mathbf{k}_0)$. We now outline its properties. Let A, B and C be filtered chain complexes over $(\mathbf{k}, \mathbf{k}_0)$. We want to sketch a proof of the fact that there is a filtered chain complex $[B,C]_{(\mathbf{k},\mathbf{k}_0)}$ over $(\mathbf{k},\mathbf{k}_0)$ such that

$$\hom_{\mathcal{F}\mathbf{Ch}(\mathbf{k},\mathbf{k}_0)}(A\otimes_{(\mathbf{k},\mathbf{k}_0)} B, C) \cong \hom_{\mathcal{F}\mathbf{Ch}(\mathbf{k},\mathbf{k}_0)}(A,[B,C]_{(\mathbf{k},\mathbf{k}_0)}).$$

We recall from §2.3.1 that given two chain complexes B' and C' over the ring $\mathbf{k}_0$, the chain complex over $\mathbf{k}_0$ of internal morphism $[B',C']_{\mathbf{k}_0}$ between them consists of $\mathbf{k}_0$-linear graded maps. Let now $f\colon A\otimes_{(\mathbf{k},\mathbf{k}_0)} B\to C$ be a morphism in $\mathcal{F}\mathbf{Ch}(\mathbf{k},\mathbf{k}_0)$. Then f restricts to chain maps of chain complexes over $\mathbf{k}_0$

$$f^\alpha\colon (A\otimes_{(\mathbf{k},\mathbf{k}_0)})B)^{\leq\alpha}\to C^{\leq\alpha}$$

for any $\alpha\in\mathbb{R}$. Up to inclusion, f^α further restricts to chain maps

$$A^{\leq\alpha_1}\otimes_{\mathbf{k}_0} B^{\leq\alpha-\alpha_1}\to C^{\leq\alpha}$$

which can be identified with graded $\mathbf{k}_0$-linear maps

$$A^{\leq\alpha_1}\to [B^{\leq\alpha-\alpha_1}, C^{\leq\alpha}]_{\mathbf{k}_0}.$$

We introduce the following definition.

Definition 2.3.5.7. Let (C,d_C) and (D,d_D) be filtered chain complexes over $(\mathbf{k}_0,\mathbf{k})$. A $\mathbf{k}$-linear graded map $f\colon C\to D$ is said to be of finite shift if there is $r\geq 0$ such that f restricts to k_0-linear graded

$$f^{\leq\alpha}:=f|_{C^{\leq\alpha}}\colon C^{\leq\alpha}\to D^{\leq\alpha+r}$$

for any $\alpha\in\mathbb{R}$. In the above case we say that f shifts filtration by $\leq r$.

From our discussion above, it follows that $[B,C]_{(\mathbf{k},\mathbf{k}_0)}$ is defined as the space of graded $\mathbf{k}$-linear maps $f\colon B\to C$ of finite shift, endowed with the usual differential of graded linear maps between chain complexes. Moreover, the filtration is defined as

$$[B,C]^{\leq r}_{(\mathbf{k},\mathbf{k}_0)}:=\left\{f\in[B,C]_{(\mathbf{k},\mathbf{k}_0)}\mid f \text{ shifts filtration by } \leq\alpha\right\}.$$

The filtered dg-category $\mathcal{F}\mathbf{Ch}_{\mathcal{F}dg}(\mathbf{k},\mathbf{k}_0)$ will be called the filtered dg-category of filtered chain complexes over $(\mathbf{k},\mathbf{k}_0)$. We will write $[B,C]_{(\mathbf{k},\mathbf{k}_0)}$ as $\hom_{\mathcal{F}\mathbf{Ch}_{\mathcal{F}dg}(\mathbf{k},\mathbf{k}_0)}(B,C)$.

Consider now the category $\mathcal{F}dg(\mathbf{k},\mathbf{k}_0)$ of filtered dg-categories over $(\mathbf{k},\mathbf{k}_0)$.

Definition 2.3.5.8. Let A and B be filtered dg-categories over $(\mathbf{k},\mathbf{k}_0)$. An enriched functor $F\colon A\to B$ is called a filtered dg-functor.

In other words, a filtered dg-functor $F\colon A\to B$ is simply a dg-functor such that the induced maps

$$F\colon \hom_A(L_0,L_1)\to\hom_B(FL_0,FL_1)$$

are filtered chain maps.

Definition 2.3.5.9. Let $F,G\colon A\to B$ be filtered dg-functors. An enriched natural transformation $\eta\colon F\to G$ is called a filtered dg-natural transformation.

In other words, a filtered natural transformation $\eta\colon F \to G$ between filtered dg-functors is a collection of morphisms $\eta_L \in \hom^{\leq 0}(FL, GL)$ of degree zero and closed.

The following is a consequence of Lemma 2.3.5.5.

DEFINITION 2.3.5.10. The category $\mathcal{F}dg(\mathbf{k}, \mathbf{k}_0)$ of filtered dg-categories over $(\mathbf{k}, \mathbf{k}_0)$ is a closed monoidal category with monoidal functor $\otimes_{\mathcal{F}dg}$ induced by the monoidal structure of $\mathcal{F}\mathbf{Ch}_{dg}(R)$ and unit given by the filtered chain complex $\mathbf{k}$ seen as a filtereddg-category with a single object and trivial differential.

Given two filtered dg-categories B and C, the dg-category of internal morphisms $[B, C]_{\mathcal{F}dg}$ appearing in the isomorphisms

$$\hom_{\mathcal{F}dg(\mathbf{k},\mathbf{k}_0)}(A \otimes B, C) \cong hom_{\mathcal{F}dg(\mathbf{k},\mathbf{k}_0)}(A, [B, C]_{\mathcal{F}dg})$$

has the same objects of $\hom_{\mathcal{F}dg(\mathbf{k},\mathbf{k}_0)}(B, C)$ (that is, filtered dg-functors $B \to C$). Let $F, G\colon B \to C$ be filtered dg-functors, then the filtered chain complex $\hom_{[B,C]_{\mathcal{F}dg}}(F, G)$ consists of graded natural transformations $\eta\colon F \to G$ such that there is $\alpha \in \mathbb{R}$ such that η_Lß $\hom_C^{\leq\alpha}(FL, GL)$ for any object L of B. As in the unfiltered case, the differential is induced object-wise by the differential on morphism spaces in the dg-category C. The filtration $(\hom^{\alpha}_{[B,C]_{\mathcal{F}dg}}(F, G)))_{\alpha\in\mathbb{R}}$ is defined in the obvious way, and is preserved by the above defined differential since C is a filtered dg-category.

2.3.6. Shift functors. We look now at the case $(\mathbf{k}, \mathbf{k}_0) = (\Lambda, \Lambda_0)$, where Λ is the Novikov field over $\mathbb{Z}_2$. The case of $\mathbb{Z}_2$ iself may be treated ad hoc or by seeing $\mathbb{Z}_2$ as $\varprojlim_{n\to\infty} \Lambda_0/T^{1/n}\Lambda_0$.

Let $\alpha \in \mathbb{R}$. We define $\Sigma^\alpha\Lambda$ to be the filtered chain complex Λ over Λ_0 filtered by

$$(\Sigma^\alpha\Lambda) := T^{\alpha-\beta}\Lambda_0.$$

We claim that the functor

$$\Sigma^\alpha\Lambda \otimes_{(\Lambda,\Lambda_0)} -\colon \mathcal{F}\mathbf{Ch}(\Lambda, \Lambda_0) \to \mathcal{F}\mathbf{Ch}(\Lambda, \Lambda_0)$$

is lax monoidal. This follows easily by considering the maps of filtered chain complexes

$$\phi\colon \Lambda \to \Sigma^\alpha\Lambda$$

defined via multiplication by T^α and

$$\phi_{A,B}\colon (\Sigma^\alpha\Lambda \otimes_{(\Lambda,\Lambda_0)} A) \otimes (\Sigma^\alpha\Lambda \otimes_{(\Lambda,\Lambda_0)} B) \to \Sigma^\alpha\Lambda \otimes_{(\Lambda,\Lambda_0)} (A \otimes_{(\Lambda,\Lambda_0)} B)$$

defined via multiplication by $T^{-\alpha}$. Equivalently, the two map above give $\Sigma^\alpha\Lambda$ the structure of monoidal object in $\mathcal{F}\mathbf{Ch}(\Lambda, \Lambda_0)$. We denote

$$\Sigma^\alpha := \Sigma^\alpha\Lambda \otimes_{(\Lambda,\Lambda_0)} -$$

and call it the α-shift functor on $\mathcal{F}\mathbf{Ch}(\Lambda, \Lambda_0)$.

Moreover, the family $(\Sigma^\alpha)_{\alpha\in\mathbb{R}}$ packs into a monoidal functor

$$\Sigma\colon \mathbb{R} \to \hom_{\mathbf{Cat}}(\mathcal{F}\mathbf{Ch}(\Lambda, \Lambda_0), \mathcal{F}\mathbf{Ch}(\Lambda, \Lambda_0)),$$

where **Cat** is the bicategory of categories and $\hom_{\mathbf{Cat}}(\mathcal{F}\mathbf{Ch}(\Lambda, \Lambda_0), \mathcal{F}\mathbf{Ch}(\Lambda, \Lambda_0))$ is the category of endofunctors $\mathcal{F}\mathbf{Ch}(\Lambda, \Lambda_0) \to \mathcal{F}\mathbf{Ch}(\Lambda, \Lambda_0)$ equipped with the usual monoidal structure induced by composition. In other words, Σ assigns to any $\alpha \in \mathbb{R}$ an endofunctor Σ^α of $\mathcal{F}\mathbf{Ch}(\Lambda, \Lambda_0)$ and satisfies $\Sigma^\alpha \circ \Sigma^\beta = \Sigma^{\alpha+\beta}$ and $\Sigma^0 = id_{\mathcal{F}\mathbf{Ch}(\Lambda,\Lambda_0)}$, and to each (unique morphism) $x \to y$ a natural transformation $\eta_{y,x} := \Sigma(x \leq y)$.

Since Σ^α is induced by the monoidal product $\otimes_{(\Lambda,\Lambda_0)}$ it induces a filtered dg-functor

$$\Sigma^\alpha \colon \mathcal{F}\mathbf{Ch}_{\mathcal{F}dg}(\Lambda, \Lambda_0) \to \mathcal{F}\mathbf{Ch}_{\mathcal{F}dg}(\Lambda, \Lambda_0).$$

These functors pack together into a monoidal functor

$$\Sigma \colon \mathbb{R} \to \hom_{dg(\Lambda,\Lambda_0)}(\mathcal{F}\mathbf{Ch}_{\mathcal{F}dg}(\Lambda, \Lambda_0), \mathcal{F}\mathbf{Ch}_{\mathcal{F}dg}(\Lambda, \Lambda_0)).$$

It is an exercise to see that Σ lifts to a monoidal functor

$$\Sigma \colon \Lambda \to \hom_{dg(\Lambda,\Lambda_0)}(\mathcal{F}\mathbf{Ch}_{\mathcal{F}dg}(\Lambda, \Lambda_0), \mathcal{F}\mathbf{Ch}_{\mathcal{F}dg}(\Lambda, \Lambda_0))$$

defined via $\Sigma(T^\alpha) := \Sigma^\alpha$ for any $\alpha \in \mathbb{R}$ and extended by linearity. Viewed in this form, Σ is actually a filtered dg-functor, in the sense that $\eta_{\alpha,\beta}$ (which, in this formalism corresponds to the image to multiplication by $T^{\alpha-\beta}$) is a natural transformation of shift $\beta - \alpha$. Note that, trivially, it is a closed natural transformation between the filtered dg-functors Σ^α and Σ^β.

DEFINITION 2.3.6.1. Let A be a filtered dg-category over (Λ, Λ_0). A shift functor on A is a monoidal filtered dg-functor

$$\Sigma \colon \Lambda \to \hom_{\mathcal{F}dg(\Lambda,\Lambda_0)}(A, A).$$

The same definition makes sense for dg-categories over $(\mathbb{Z}_2, \mathbb{Z}_2)$.

The following result is a rewriting of Remark 2.16 in [**BCZ24**].

LEMMA 2.3.6.2. *Let A be a filtered dg-category with shift functor Σ and let $\alpha \in \mathbb{R}$. Then for all objects L_0 and L_1 of A we have*

$$\hom_A(\Sigma^\alpha L_0, L_1) \cong \Sigma^\alpha \hom_A(L_0, L_1)$$

where the first Σ denotes the shift functor on A, while the second denotes the shift functor on $\mathcal{F}\mathbf{Ch}_{\mathcal{F}dg}(\Lambda, \Lambda_0)$.

2.4. A dry introduction to filtered A_∞-categories

The material in this section is an adaptation of Appendix A in [**ABC26**] and Section 1.1 in [**Amb25**].

A_∞-categories (over a ring R) are homotopical refinement of dg-categories, and can be tought of as categories "homotopically" enriched over chain complexes over R. More precisely, an A_∞-category is a category enriched over a certain operad of cellular complexes in the category of chain complexes over R (see, for instance, [**AB24**]). This viewpoint mirrors the analogous one we described for dg-categories in §2.3, and we plan to study it in depth in future work. In §2.3 we introduced several notions that will have an A_∞-counterpart in the following.

The A_∞-versions introduced here will not be based on a categorical tratment, but purely based on formulae - hence the "dry" in the title of this section.

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